Evaluation runs: **reset** (s5-reset)

generated 2026-11-08 from brain-data 8a2534d6f 2026-11-08 20:14:59 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): reset: expected violation; ctl: expected violation (the matched control is a violating one); twin: expected conforms. Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". G_{ref}, G_{cand} = the graphs of the reference and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is G_{ref}^{expl}); G_x = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

stateless

stateless: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G_{ref}^{expl}	run
1,1	1	1	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
1,3	6	6	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
1,4	24	24	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
2,1	1	1	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
2,3	6	6	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
2,4	24	24	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
3,1	1	1	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
3,3	6	6	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
3,4	24	24	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
4,1	1	1	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
4,3	6	6	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
4,4	24	24	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
5,1	1	1	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
5,3	6	6	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
5,4	24	24	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
6,1	1	1	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
6,3	6	6	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
6,4	24	24	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
7,1	1	1	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
7,3	6	6	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
7,4	24	24	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
8,1	1	1	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
8,3	6	6	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
8,4	24	24	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
9,1	1	1	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
9,3	6	6	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
9,4	24	24	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
10,1	1	1	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
10,3	6	6	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
10,4	24	24	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
10,5	120	120	conforms	0.1	2	1 h, 250 GB	1	1	brain01, 2026-11-02
10,6	720	720	conforms	0.6	2	1 h, 250 GB	1	1	brain01, 2026-11-02
10,7	5040	5040	conforms	4.3	2	1 h, 250 GB	1	1	brain01, 2026-11-02
10,8	40320	40320	conforms	36.3	10	1 h, 250 GB	1	1	brain01, 2026-11-02
10,9	362880	362880	conforms	336.5	10	1 h, 250 GB	1	1	brain01, 2026-11-02
10,10	3628800	3628800	conforms	3500.3	10	1 h, 250 GB	1	1	brain01, 2026-11-02

stateless: pair **ctl**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	sweeps	G_{ref}^{expl}	run
1,1	1	2	1	violation	0.0	2	1 h, 250 GB	2	2	brain03, 2026-11-02
1,3	6	2	1	violation	0.0	4	1 h, 250 GB	2	7	brain03, 2026-11-02
1,4	24	2	1	violation	0.0	2	1 h, 250 GB	2	25	brain03, 2026-11-02
2,1	1	4	3	violation	0.0	2	1 h, 250 GB	4	4	brain03, 2026-11-02
2,3	6	4	3	violation	0.0	2	1 h, 250 GB	4	19	brain03, 2026-11-02
2,4	24	4	3	violation	0.0	2	1 h, 250 GB	4	73	brain03, 2026-11-02
3,1	1	8	7	violation	0.0	2	1 h, 250 GB	8	8	brain03, 2026-11-02
3,3	6	8	7	violation	0.0	2	1 h, 250 GB	8	43	brain03, 2026-11-02
3,4	24	8	7	violation	0.1	2	1 h, 250 GB	8	169	brain03, 2026-11-02
4,1	1	16	15	violation	0.0	2	1 h, 250 GB	16	16	brain03, 2026-11-02
4,3	6	16	15	violation	0.0	2	1 h, 250 GB	16	91	brain03, 2026-11-02
4,4	24	16	15	violation	0.1	2	1 h, 250 GB	16	361	brain03, 2026-11-02
5,1	1	32	31	violation	0.0	2	1 h, 250 GB	32	32	brain03, 2026-11-02
5,3	6	32	31	violation	0.1	2	1 h, 250 GB	32	187	brain03, 2026-11-02
5,4	24	32	31	violation	0.1	2	1 h, 250 GB	32	745	brain03, 2026-11-02
6,1	1	64	63	violation	0.1	2	1 h, 250 GB	64	64	brain03, 2026-11-02
6,3	6	64	63	violation	0.1	2	1 h, 250 GB	64	379	brain03, 2026-11-02
6,4	24	64	63	violation	0.2	2	1 h, 250 GB	64	1513	brain03, 2026-11-02
7,1	1	128	127	violation	0.1	4	1 h, 250 GB	128	128	brain03, 2026-11-02
7,3	6	128	127	violation	0.2	4	1 h, 250 GB	128	763	brain03, 2026-11-02
7,4	24	128	127	violation	0.5	4	1 h, 250 GB	128	3049	brain03, 2026-11-02
8,1	1	256	255	violation	0.3	8	1 h, 250 GB	256	256	brain03, 2026-11-02
8,3	6	256	255	violation	0.5	8	1 h, 250 GB	256	1531	brain03, 2026-11-02
8,4	24	256	255	violation	1.1	6	1 h, 250 GB	256	6121	brain03, 2026-11-02
9,1	1	512	511	violation	0.7	16	1 h, 250 GB	512	512	brain03, 2026-11-02
9,3	6	512	511	violation	1.1	16	1 h, 250 GB	512	3067	brain03, 2026-11-02
9,4	24	512	511	violation	2.8	12	1 h, 250 GB	512	12265	brain03, 2026-11-02
10,1	1	1024	1023	violation	1.8	32	1 h, 250 GB	1024	1024	brain03, 2026-11-02
10,3	6	1024	1023	violation	2.8	30	1 h, 250 GB	1024	6139	brain03, 2026-11-02
10,4	24	1024	1023	violation	6.4	26	1 h, 250 GB	1024	24553	brain03, 2026-11-02
10,5	120	1024	1023	violation	27.4	30	1 h, 250 GB	1024	122761	brain01, 2026-11-02
10,6	720	1024	1023	violation	156.1	30	1 h, 250 GB	1024	736561	brain01, 2026-11-02
10,7	5040	1024	1023	violation	1101.9	24	1 h, 250 GB	1024	5155921	brain01, 2026-11-02
10,8	40320	1024	1023	violation	9207.0	26	48 h, 250 GB	1024	41247361	brain01, 2026-11-03

stateless: pair **reset**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	sweeps	G_{ref}^{expl}	run
1,1	1	2	1	violation	0.0	2	1 h, 250 GB	2	2	brain03, 2026-11-02
1,3	6	2	1	violation	0.0	2	1 h, 250 GB	2	7	brain03, 2026-11-02
1,4	24	2	1	violation	0.0	2	1 h, 250 GB	2	25	brain03, 2026-11-02
2,1	1	4	3	violation	0.0	2	1 h, 250 GB	4	4	brain03, 2026-11-02
2,3	6	4	3	violation	0.0	2	1 h, 250 GB	4	19	brain03, 2026-11-02
2,4	24	4	3	violation	0.0	2	1 h, 250 GB	4	73	brain03, 2026-11-02
3,1	1	8	7	violation	0.0	2	1 h, 250 GB	8	8	brain03, 2026-11-02
3,3	6	8	7	violation	0.0	2	1 h, 250 GB	8	43	brain03, 2026-11-02
3,4	24	8	7	violation	0.0	2	1 h, 250 GB	8	169	brain03, 2026-11-02
4,1	1	16	15	violation	0.0	2	1 h, 250 GB	16	16	brain03, 2026-11-02
4,3	6	16	15	violation	0.0	2	1 h, 250 GB	16	91	brain03, 2026-11-02
4,4	24	16	15	violation	0.1	2	1 h, 250 GB	16	361	brain03, 2026-11-02
5,1	1	32	31	violation	0.0	2	1 h, 250 GB	32	32	brain03, 2026-11-02
5,3	6	32	31	violation	0.1	2	1 h, 250 GB	32	187	brain03, 2026-11-02
5,4	24	32	31	violation	0.1	2	1 h, 250 GB	32	745	brain03, 2026-11-02
6,1	1	64	63	violation	0.1	2	1 h, 250 GB	64	64	brain03, 2026-11-02
6,3	6	64	63	violation	0.1	2	1 h, 250 GB	64	379	brain03, 2026-11-02
6,4	24	64	63	violation	0.2	2	1 h, 250 GB	64	1513	brain03, 2026-11-02
7,1	1	128	127	violation	0.1	4	1 h, 250 GB	128	128	brain03, 2026-11-02
7,3	6	128	127	violation	0.3	6	1 h, 250 GB	128	763	brain03, 2026-11-02
7,4	24	128	127	violation	0.5	4	1 h, 250 GB	128	3049	brain03, 2026-11-02
8,1	1	256	255	violation	0.3	10	1 h, 250 GB	256	256	brain03, 2026-11-02
8,3	6	256	255	violation	0.5	8	1 h, 250 GB	256	1531	brain03, 2026-11-02

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	sweeps	G_{ref}^{expl}	run
8,4	24	256	255	violation	1.1	8	1 h, 250 GB	256	6121	brain03, 2026-11-02
9,1	1	512	511	violation	0.7	16	1 h, 250 GB	512	512	brain03, 2026-11-02
9,3	6	512	511	violation	1.0	18	1 h, 250 GB	512	3067	brain03, 2026-11-02
9,4	24	512	511	violation	2.4	16	1 h, 250 GB	512	12265	brain03, 2026-11-02
10,1	1	1024	1023	violation	1.2	30	1 h, 250 GB	1024	1024	brain03, 2026-11-02
10,3	6	1024	1023	violation	2.0	30	1 h, 250 GB	1024	6139	brain03, 2026-11-02
10,4	24	1024	1023	violation	4.6	34	1 h, 250 GB	1024	24553	brain03, 2026-11-02
10,5	120	1024	1023	violation	19.6	30	1 h, 250 GB	1024	122761	brain01, 2026-11-02
10,6	720	1024	1023	violation	113.2	34	1 h, 250 GB	1024	736561	brain01, 2026-11-02
10,7	5040	1024	1023	violation	801.4	30	1 h, 250 GB	1024	5155921	brain01, 2026-11-02
10,8	40320	1024	1023	violation	6572.0	30	48 h, 250 GB	1024	41247361	brain01, 2026-11-03

Not run (stateless). planned but never scheduled (grid points beyond the ladder): twin 10 8 bag, ctl 10 8 bag, reset 10 8 bag, ctl 10 9 bag, reset 10 9 bag, ctl 10 10 bag, reset 10 10 bag, reset 11 1 bag, ctl 11 1 bag, twin 11 1 bag, twin 11 3 bag, reset 11 3 bag, ctl 11 3 bag, twin 11 4 bag, reset 11 4 bag, ctl 11 4 bag, ctl 12 1 bag, reset 12 1 bag, twin 12 1 bag, twin 12 3 bag, reset 12 3 bag, ctl 12 3 bag, twin 12 4 bag, ctl 12 4 bag, … (38 instances in all).

left out by the table rules (invalid: foreign load): reset 20 11 bag.

stateful

stateful: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
1,1	1	1	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
1,3	6	6	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
1,4	24	24	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
2,1	1	1	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
2,3	6	6	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
2,4	24	24	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
3,1	1	1	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
3,3	6	6	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
3,4	24	24	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
4,1	1	1	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
4,3	6	6	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
4,4	24	24	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
5,1	1	1	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
5,3	6	6	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
5,4	24	24	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
6,1	1	1	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
6,3	6	6	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
6,4	24	24	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
7,1	1	1	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
7,3	6	6	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
7,4	24	24	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
8,1	1	1	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
8,3	6	6	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
8,4	24	24	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
9,1	1	1	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
9,3	6	6	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
9,4	24	24	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
10,1	1	1	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
10,3	6	6	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
10,4	24	24	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
10,5	120	120	conforms	0.1	2	1 h, 250 GB	brain01, 2026-11-02
10,6	720	720	conforms	0.3	2	1 h, 250 GB	brain01, 2026-11-02
10,7	5040	5040	conforms	2.1	2	1 h, 250 GB	brain01, 2026-11-02
10,8	40320	40320	conforms	17.1	10	1 h, 250 GB	brain01, 2026-11-02
10,9	362880	362880	conforms	157.8	10	1 h, 250 GB	brain01, 2026-11-02
10,10	3628800	3628800	conforms	1601.8	10	1 h, 250 GB	brain01, 2026-11-02

stateful: pair **ctl**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	run
1,1	1	2	1	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
1,3	6	2	1	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
1,4	24	2	1	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
2,1	1	4	3	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
2,3	6	4	3	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
2,4	24	4	3	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
3,1	1	8	7	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
3,3	6	8	7	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
3,4	24	8	7	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
4,1	1	16	15	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
4,3	6	16	15	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
4,4	24	16	15	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
5,1	1	32	31	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
5,3	6	32	31	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
5,4	24	32	31	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
6,1	1	64	63	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
6,3	6	64	63	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
6,4	24	64	63	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
7,1	1	128	127	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
7,3	6	128	127	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
7,4	24	128	127	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
8,1	1	256	255	violation	0.1	8	1 h, 250 GB	brain02, 2026-11-02
8,3	6	256	255	violation	0.1	8	1 h, 250 GB	brain02, 2026-11-02
8,4	24	256	255	violation	0.1	8	1 h, 250 GB	brain02, 2026-11-02
9,1	1	512	511	violation	0.2	14	1 h, 250 GB	brain02, 2026-11-02
9,3	6	512	511	violation	0.3	14	1 h, 250 GB	brain02, 2026-11-02
9,4	24	512	511	violation	0.3	14	1 h, 250 GB	brain02, 2026-11-02
10,1	1	1024	1023	violation	0.8	34	1 h, 250 GB	brain02, 2026-11-02
10,3	6	1024	1023	violation	0.8	36	1 h, 250 GB	brain02, 2026-11-02
10,4	24	1024	1023	violation	0.8	36	1 h, 250 GB	brain02, 2026-11-02
10,5	120	1024	1023	violation	0.8	34	1 h, 250 GB	brain01, 2026-11-02
10,6	720	1024	1023	violation	0.9	34	1 h, 250 GB	brain01, 2026-11-02
10,7	5040	1024	1023	violation	1.8	34	1 h, 250 GB	brain01, 2026-11-02
10,8	40320	1024	1023	violation	9.0	34	1 h, 250 GB	brain01, 2026-11-02
10,9	362880	1024	1023	violation	77.5	34	1 h, 250 GB	brain01, 2026-11-02
10,10	3628800	1024	1023	violation	791.0	34	1 h, 250 GB	brain01, 2026-11-02

stateful: pair **reset**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	run
1,1	1	2	1	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
1,3	6	2	1	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
1,4	24	2	1	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
2,1	1	4	3	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
2,3	6	4	3	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
2,4	24	4	3	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
3,1	1	8	7	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
3,3	6	8	7	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
3,4	24	8	7	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
4,1	1	16	15	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
4,3	6	16	15	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
4,4	24	16	15	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
5,1	1	32	31	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
5,3	6	32	31	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
5,4	24	32	31	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
6,1	1	64	63	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
6,3	6	64	63	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
6,4	24	64	63	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
7,1	1	128	127	violation	0.1	4	1 h, 250 GB	brain02, 2026-11-02
7,3	6	128	127	violation	0.1	4	1 h, 250 GB	brain02, 2026-11-02
7,4	24	128	127	violation	0.1	4	1 h, 250 GB	brain02, 2026-11-02
8,1	1	256	255	violation	0.1	8	1 h, 250 GB	brain02, 2026-11-02
8,3	6	256	255	violation	0.1	6	1 h, 250 GB	brain02, 2026-11-02
8,4	24	256	255	violation	0.1	8	1 h, 250 GB	brain02, 2026-11-02

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	run
9,1	1	512	511	violation	0.2	14	1 h, 250 GB	brain02, 2026-11-02
9,3	6	512	511	violation	0.2	18	1 h, 250 GB	brain02, 2026-11-02
9,4	24	512	511	violation	0.2	16	1 h, 250 GB	brain02, 2026-11-02
10,1	1	1024	1023	violation	0.5	38	1 h, 250 GB	brain02, 2026-11-02
10,3	6	1024	1023	violation	0.5	36	1 h, 250 GB	brain02, 2026-11-02
10,4	24	1024	1023	violation	0.5	36	1 h, 250 GB	brain02, 2026-11-02
10,5	120	1024	1023	violation	0.6	34	1 h, 250 GB	brain01, 2026-11-02
10,6	720	1024	1023	violation	0.6	36	1 h, 250 GB	brain01, 2026-11-02
10,7	5040	1024	1023	violation	1.6	32	1 h, 250 GB	brain01, 2026-11-02
10,8	40320	1024	1023	violation	9.0	34	1 h, 250 GB	brain01, 2026-11-02
10,9	362880	1024	1023	violation	79.8	32	1 h, 250 GB	brain01, 2026-11-02
10,10	3628800	1024	1023	violation	807.5	32	1 h, 250 GB	brain01, 2026-11-02

Not run (stateful). planned but never scheduled (grid points beyond the ladder): twin 10 8 bag, ctl 10 8 bag, reset 10 8 bag, reset 11 1 bag, ctl 11 1 bag, twin 11 1 bag, twin 11 3 bag, reset 11 3 bag, ctl 11 3 bag, twin 11 4 bag, reset 11 4 bag, ctl 11 4 bag, ctl 12 1 bag, reset 12 1 bag, twin 12 1 bag, twin 12 3 bag, reset 12 3 bag, ctl 12 3 bag, twin 12 4 bag, ctl 12 4 bag, reset 12 4 bag, reset 13 1 bag, ctl 13 1 bag, twin 13 1 bag, … (34 instances in all).

the tool ended with an error (exit-101): reset 20 11 bag.

offline

offline: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1	1	1	conforms	0.0	8	10 min, 16 GB	0.0	0.0	0.0	in-memory	brain01, 2026-11-02
1,3	6	6	conforms	0.1	8	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain01, 2026-11-02
1,4	24	24	conforms	0.1	8	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain01, 2026-11-02
2,1	1	1	conforms	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
2,3	6	6	conforms	0.1	8	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain01, 2026-11-02
2,4	24	24	conforms	0.1	8	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain01, 2026-11-02
3,1	1	1	conforms	0.0	8	10 min, 16 GB	0.0	0.0	0.0	in-memory	brain01, 2026-11-02
3,3	6	6	conforms	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
3,4	24	24	conforms	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
4,1	1	1	conforms	0.1	8	10 min, 16 GB	0.1	0.0	0.0	in-memory	brain01, 2026-11-02
4,3	6	6	conforms	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
4,4	24	24	conforms	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
5,1	1	1	conforms	0.2	8	10 min, 16 GB	0.1	0.2	0.0	in-memory	brain01, 2026-11-02
5,3	6	6	conforms	0.2	8	10 min, 16 GB	0.0	0.2	0.0	in-memory	brain01, 2026-11-02
5,4	24	24	conforms	0.2	8	10 min, 16 GB	0.2	0.1	0.0	in-memory	brain01, 2026-11-02
6,1	1	1	conforms	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
6,3	6	6	conforms	0.2	8	10 min, 16 GB	0.1	0.2	0.0	in-memory	brain01, 2026-11-02
6,4	24	24	conforms	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
7,1	1	1	conforms	0.1	8	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain01, 2026-11-02
7,3	6	6	conforms	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
7,4	24	24	conforms	0.2	8	10 min, 16 GB	0.2	0.1	0.0	in-memory	brain01, 2026-11-02
8,1	1	1	conforms	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
8,3	6	6	conforms	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
8,4	24	24	conforms	0.4	8	10 min, 16 GB	0.4	0.1	0.0	in-memory	brain01, 2026-11-02
9,1	1	1	conforms	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
9,3	6	6	conforms	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
9,4	24	24	conforms	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
10,1	1	1	conforms	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
10,3	6	6	conforms	0.2	8	10 min, 16 GB	0.2	0.1	0.0	in-memory	brain01, 2026-11-02
10,4	24	24	conforms	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
10,5	120	120	conforms	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
10,6	720	720	conforms	0.3	8	10 min, 16 GB	0.3	0.2	0.0	in-memory	brain01, 2026-11-02
10,7	5040	5040	conforms	1.4	8	10 min, 16 GB	1.4	1.3	0.0	in-memory	brain01, 2026-11-02
10,8	40320	40320	conforms	10.7	19	10 min, 16 GB	10.7	10.7	0.0	in-memory	brain01, 2026-11-02
10,9	362880	362880	conforms	98.5	19	1 h, 250 GB	98.5	97.3	0.0	merge	brain01, 2026-11-03
10,10	3628800	3628800	conforms	996.2	19	1 h, 250 GB	992.2	996.2	0.0	merge	brain01, 2026-11-03

offline: pair **ctl**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1	1	2	1	violation	0.1	8	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain01, 2026-11-02
1,3	6	2	1	violation	0.1	8	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain01, 2026-11-02
1,4	24	2	1	violation	0.0	8	10 min, 16 GB	0.0	0.0	0.0	in-memory	brain01, 2026-11-02
2,1	1	4	3	violation	0.1	8	10 min, 16 GB	0.1	0.0	0.0	in-memory	brain01, 2026-11-02
2,3	6	4	3	violation	0.1	8	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain01, 2026-11-02
2,4	24	4	3	violation	0.1	8	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain01, 2026-11-02
3,1	1	8	7	violation	0.0	8	10 min, 16 GB	0.0	0.0	0.0	in-memory	brain01, 2026-11-02
3,3	6	8	7	violation	0.1	8	10 min, 16 GB	0.1	0.0	0.0	in-memory	brain01, 2026-11-02
3,4	24	8	7	violation	0.2	8	10 min, 16 GB	0.1	0.2	0.0	in-memory	brain01, 2026-11-02
4,1	1	16	15	violation	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
4,3	6	16	15	violation	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
4,4	24	16	15	violation	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
5,1	1	32	31	violation	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
5,3	6	32	31	violation	0.2	8	10 min, 16 GB	0.0	0.2	0.0	in-memory	brain01, 2026-11-02
5,4	24	32	31	violation	0.3	8	10 min, 16 GB	0.2	0.3	0.0	in-memory	brain01, 2026-11-02
6,1	1	64	63	violation	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
6,3	6	64	63	violation	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
6,4	24	64	63	violation	0.2	8	10 min, 16 GB	0.1	0.2	0.0	in-memory	brain01, 2026-11-02
7,1	1	128	127	violation	0.1	8	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain01, 2026-11-02
7,3	6	128	127	violation	0.2	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
7,4	24	128	127	violation	0.2	8	10 min, 16 GB	0.2	0.1	0.0	in-memory	brain01, 2026-11-02
8,1	1	256	255	violation	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
8,3	6	256	255	violation	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
8,4	24	256	255	violation	0.4	8	10 min, 16 GB	0.4	0.1	0.0	in-memory	brain01, 2026-11-02
9,1	1	512	511	violation	0.2	12	10 min, 16 GB	0.1	0.1	0.1	in-memory	brain01, 2026-11-02
9,3	6	512	511	violation	0.3	12	10 min, 16 GB	0.1	0.3	0.1	in-memory	brain01, 2026-11-02
9,4	24	512	511	violation	0.2	12	10 min, 16 GB	0.1	0.1	0.1	in-memory	brain01, 2026-11-02
10,1	1	1024	1023	violation	0.4	24	10 min, 16 GB	0.1	0.3	0.1	in-memory	brain01, 2026-11-02
10,3	6	1024	1023	violation	0.4	24	10 min, 16 GB	0.2	0.3	0.1	in-memory	brain01, 2026-11-02
10,4	24	1024	1023	violation	0.4	24	10 min, 16 GB	0.1	0.3	0.1	in-memory	brain01, 2026-11-02
10,5	120	1024	1023	violation	0.5	24	10 min, 16 GB	0.1	0.4	0.1	in-memory	brain01, 2026-11-02
10,6	720	1024	1023	violation	0.5	24	10 min, 16 GB	0.3	0.4	0.1	in-memory	brain01, 2026-11-02
10,7	5040	1024	1023	violation	1.5	24	10 min, 16 GB	1.4	0.3	0.1	in-memory	brain01, 2026-11-02
10,8	40320	1024	1023	violation	10.8	17	1 h, 250 GB	10.7	0.3	0.1	merge	brain01, 2026-11-03

offline: pair **reset**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1	1	2	1	violation	0.0	8	10 min, 16 GB	0.0	0.0	0.0	in-memory	brain01, 2026
1,3	6	2	1	violation	0.1	8	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain01, 2026
1,4	24	2	1	violation	0.1	8	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain01, 2026
2,1	1	4	3	violation	0.1	8	10 min, 16 GB	0.1	0.0	0.0	in-memory	brain01, 2026
2,3	6	4	3	violation	0.0	8	10 min, 16 GB	0.0	0.0	0.0	in-memory	brain01, 2026
2,4	24	4	3	violation	0.0	8	10 min, 16 GB	0.0	0.0	0.0	in-memory	brain01, 2026
3,1	1	8	7	violation	0.0	8	10 min, 16 GB	0.0	0.0	0.0	in-memory	brain01, 2026
3,3	6	8	7	violation	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026
3,4	24	8	7	violation	0.1	8	10 min, 16 GB	0.1	0.0	0.0	in-memory	brain01, 2026
4,1	1	16	15	violation	0.1	8	10 min, 16 GB	0.1	0.0	0.0	in-memory	brain01, 2026
4,3	6	16	15	violation	0.2	8	10 min, 16 GB	0.1	0.2	0.0	in-memory	brain01, 2026
4,4	24	16	15	violation	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026
5,1	1	32	31	violation	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026
5,3	6	32	31	violation	0.1	8	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain01, 2026
5,4	24	32	31	violation	0.2	8	10 min, 16 GB	0.2	0.1	0.0	in-memory	brain01, 2026
6,1	1	64	63	violation	0.2	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026
6,3	6	64	63	violation	0.2	8	10 min, 16 GB	0.1	0.2	0.0	in-memory	brain01, 2026
6,4	24	64	63	violation	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026
7,1	1	128	127	violation	0.1	8	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain01, 2026
7,3	6	128	127	violation	0.2	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026
7,4	24	128	127	violation	0.2	8	10 min, 16 GB	0.2	0.1	0.0	in-memory	brain01, 2026
8,1	1	256	255	violation	0.2	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026
8,3	6	256	255	violation	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026
8,4	24	256	255	violation	0.4	8	10 min, 16 GB	0.4	0.2	0.0	in-memory	brain01, 2026
9,1	1	512	511	violation	0.2	12	10 min, 16 GB	0.1	0.2	0.1	in-memory	brain01, 2026
9,3	6	512	511	violation	0.3	12	10 min, 16 GB	0.1	0.3	0.1	in-memory	brain01, 2026

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
9,4	24	512	511	violation	0.3	12	10 min, 16 GB	0.1	0.3	0.1	in-memory	brain01, 202
10,1	1	1024	1023	violation	0.5	24	10 min, 16 GB	0.1	0.3	0.1	in-memory	brain01, 202
10,3	6	1024	1023	violation	0.5	24	10 min, 16 GB	0.2	0.3	0.1	in-memory	brain01, 202
10,4	24	1024	1023	violation	0.4	24	10 min, 16 GB	0.1	0.3	0.1	in-memory	brain01, 202
10,5	120	1024	1023	violation	0.4	24	10 min, 16 GB	0.1	0.3	0.1	in-memory	brain01, 202
10,6	720	1024	1023	violation	0.4	24	10 min, 16 GB	0.3	0.3	0.1	in-memory	brain01, 202
10,7	5040	1024	1023	violation	1.5	24	10 min, 16 GB	1.4	0.3	0.1	in-memory	brain01, 202
10,8	40320	1024	1023	violation	11.2	17	1 h, 250 GB	10.7	0.3	0.5	merge	brain01, 202
20,11	39916800	1048576	1048575	violation	27587.4	22839	48 h, 250 GB	27257.0	103.9	330.4	merge	brain04, 202

Not run (offline). planned but never scheduled (grid points beyond the ladder): reset 11 1 bag, ctl 11 1 bag, twin 11 1 bag, twin 11 3 bag, reset 11 3 bag, ctl 11 3 bag, twin 11 4 bag, reset 11 4 bag, ctl 11 4 bag, ctl 12 1 bag, reset 12 1 bag, twin 12 1 bag, twin 12 3 bag, reset 12 3 bag, ctl 12 3 bag, twin 12 4 bag, ctl 12 4 bag, reset 12 4 bag, reset 13 1 bag, ctl 13 1 bag, twin 13 1 bag, ctl 13 3 bag, twin 13 3 bag, reset 13 3 bag, … (27 instances in all).

refused (an input missing at run time): ctl 10 9 bag, reset 10 9 bag, reset 10 10 bag, ctl 10 10 bag.